

SHP Seed Grant Cover page

Project Title: Bridging the Gap Between Clinic and Road: A Multimodal Assessment of Driving Using Mobile Health

Principal Investigator (PI):

Dr. Hannah Mercier, PhD, OTR/L

Occupational Therapy

School of Health Professions

Stony Brook University

Email: hannah.mercier@stonybrook.edu

Google Scholar Profile: <https://scholar.google.com/citations?user=IVXn63cAAAAJ&hl=en>

CO-Investigator (CO-PI)

Dr. Rosalia Casanova, OTD, OTR/L, DRP, CDI

Occupational Therapy

School of Health Professions

Stony Brook University

Email: rosalia.casanova@stonybrook.edu

Total Budget Requested:

\$5,000

Project Summary

Older adults face increased challenges with driving and community mobility due to natural aging or conditions. Despite these needs, many do not receive appropriate rehabilitative services. This proposed study builds on a pilot project led by the PI team to develop a mobile health application designed to enhance driver assessment.

The current study aims to advance the app's capabilities to evaluate real-world driving behaviors alongside standardized clinical assessments. Older drivers and their passenger caregivers will use the app to log real-time performance and safety observations, extending the scope of traditional clinic-based evaluations. This will generate objective insights from naturalistic driving experiences to inform drivers, caregivers, and providers about safety and functional needs.

The PI, an expert in mobile health and technology, will lead development to improve accessibility and usability across care settings. The Co-PI, a certified Driving Rehabilitation Professional and NYS Certified Driving Instructor, will oversee clinical assessments and translate results into practical, real-world recommendations. Collaborating with the Department Chair of Neurology, this interdisciplinary effort will establish a foundation for expanding driver rehabilitation services in Long Island.

This project aligns with SHP Seed Grant priorities: healthcare innovation, interdisciplinary collaboration, and technology integration for underserved populations.